

2012 RI H2 Mathematics Preliminary Examination Paper 1

Qn	Solution
1 [3]	$x + y + z = 21$ ----- (1) $3x - z = 0$ ----- (2) $0.05y + 0.8z = 3(0.9x) \Rightarrow 2.7x - 0.05y - 0.8z = 0$ ----- (3) Using GC, $x = 2.1, y = 12.6$ and $z = 6.3$
2 [2] [3]	$\frac{1}{(1+2x^2)^2} = (1+2x^2)^{-2}$ $= 1 + (-2)(2x^2) + \frac{(-2)(-3)}{2!}(2x^2)^2 + \frac{(-2)(-3)(-4)}{3!}(2x^2)^3 + \dots$ $= 1 - 4x^2 + 12x^4 - 32x^6 + \dots$ Coefficient of x^{2r} is $\frac{(-2)(-3)\dots(-r-1)}{r!}2^r = \frac{(-1)^r(r+1)!}{r!}2^r = (-1)^r(r+1)2^r$ Expansion is valid for $2x^2 < 1$ $\Rightarrow x^2 < \frac{1}{2}$ $\Rightarrow -\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$ $\therefore$ Range of validity is $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.
3 [6]	$w^2 = a^2 - b^2 + 2abi = 5 + 12i$ $a^2 - b^2 = 5$ (1) $2ab = 12$ (2) Sub $a = \frac{6}{b}$ in (1) $\left(\frac{6}{b}\right)^2 - b^2 = 5 \Rightarrow (b^2)^2 + 5(b^2) - 36 = 0$ $\Rightarrow (b^2 + 9)(b^2 - 4) = 0$ $\Rightarrow b = 2$ or -2 $b = 2, a = 3$ or $b = -2, a = -3$ $w = 3 + 2i$ or $-(3 + 2i)$
	$z^2 - z - (1 + 3i) = 0$ $z = \frac{1 \pm \sqrt{1 + 4(1 + 3i)}}{2} = \frac{1 \pm \sqrt{5 + 12i}}{2}$ $= \frac{1 \pm (3 + 2i)}{2}$ $z = 2 + i$ or $-1 - i$

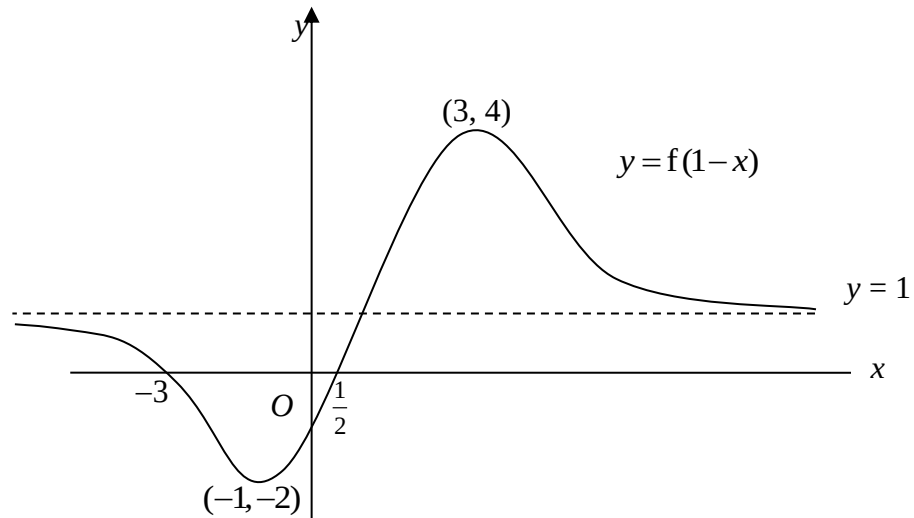
[illegible]

Alternatively,

$$\begin{aligned}
 & \int_1^4 \frac{1}{1+2\sqrt{x}+x} dx \\
 &= \int_1^2 \frac{1}{1+2u+u^2} 2u du \\
 &= \int_1^2 \frac{2u}{(1+u)^2} du \\
 &= 2 \int_1^2 \frac{1}{1+u} - \frac{1}{(1+u)^2} du \\
 &= 2 \left[\ln(1+u) + \frac{1}{1+u} \right]_1^2 \\
 &= 2 \left(\ln 3 + \frac{1}{3} \right) - 2 \left(\ln 2 + \frac{1}{2} \right) \\
 &= 2 \ln \frac{3}{2} - \frac{1}{3}
 \end{aligned}$$

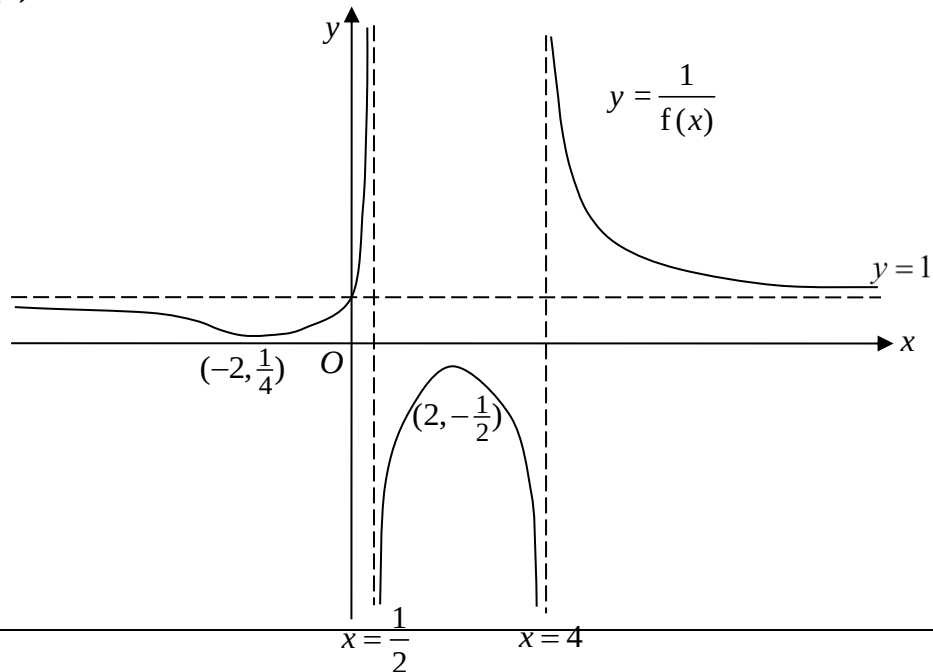
6
[3]

(i)



[3]

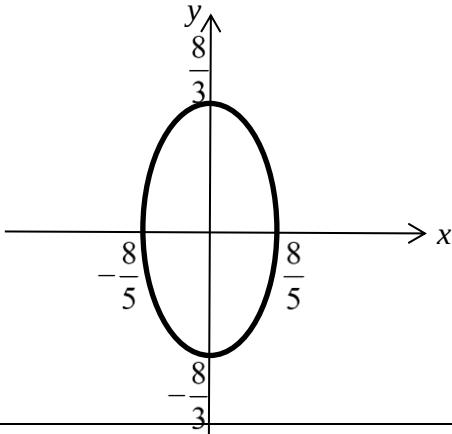
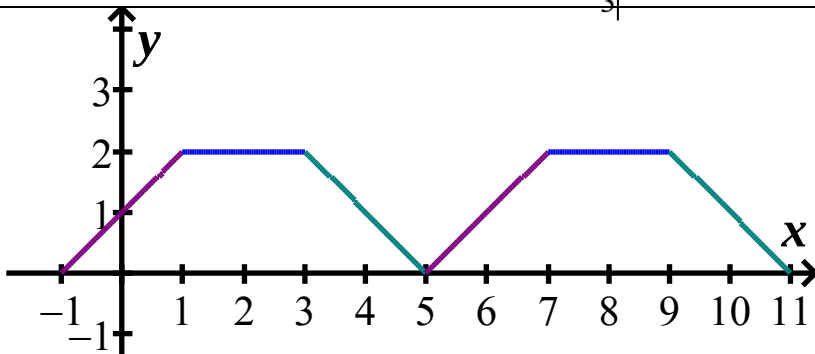
(ii)

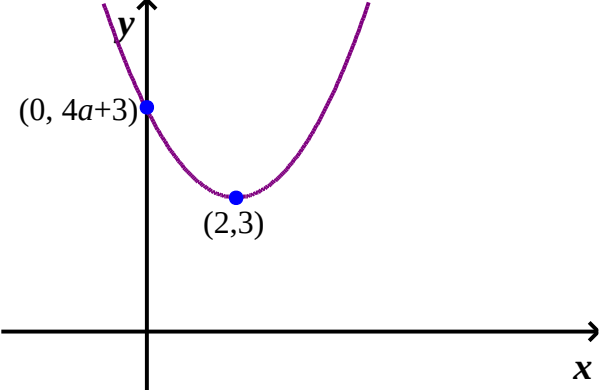
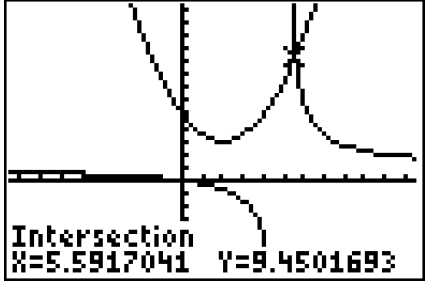


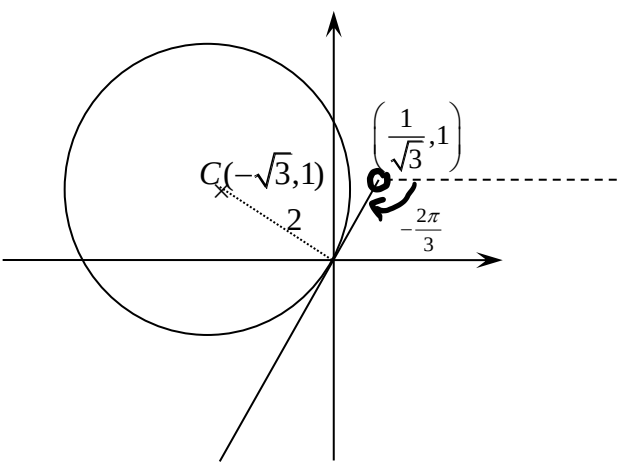
7(a) [3]	$\frac{PR}{RQ} = \frac{1}{3}, \quad \overrightarrow{OR} = \frac{1}{4}(3\overrightarrow{OP} + \overrightarrow{OQ})$ $= \frac{3}{4}(\underline{a} + \underline{b}) + \frac{1}{4}(3\underline{a} - 3\underline{b})$ $= \frac{3}{2}\underline{a} = \frac{3}{2}\overrightarrow{OA}$ $\therefore \overrightarrow{OR} // \overrightarrow{OA}$ and O is a common point O, A, R are collinear
7bi [2]	From GC $x = z$ $y = \frac{3}{2}z$ $z = z$ Cartesian equation is $\frac{x}{2} = \frac{y}{3} = \frac{z}{2}$
7bii [2]	Any point (x, y, z) on l has $x = \lambda$, $y = \frac{3\lambda}{2}$ and $z = \lambda$ for some real λ. So $(x - 2y + 2z) + c(2x - 2y + z)$ $= \left(\lambda - 2\left(\frac{3\lambda}{2}\right) + 2\lambda \right) + c \left(2\lambda - 2\left(\frac{3\lambda}{2}\right) + \lambda \right)$ $= 0 \text{ for all } c$ Hence l lies in p_3 for all c. OR All points (x, y, z) on l satisfy the equations $x - 2y + 2z = 0$ and $2x - 2y + z = 0$. So $(x - 2y + 2z) + c(2x - 2y + z) = 0 + c(0) = 0$. Hence l lies in p_3 for all c.
biii [1]	$d \neq 0$
8(i) [5]	$y = \tan^{-1}(x)$ At $x = -1$, $y = \tan^{-1}(-1) = -\frac{\pi}{4}$ and $y = -2(-1) - 2 - \frac{\pi}{4} = -\frac{\pi}{4}$

	Area of region $R = \frac{1}{2} \left(\frac{\pi}{8} \right) \left(\frac{\pi}{4} \right) + \int_{-1}^0 (0 - \tan^{-1}(x)) dx$ $= \frac{\pi^2}{64} - \left\{ \left[x \tan^{-1}(x) \right]_{-1}^0 - \int_{-1}^0 \frac{x}{1+x^2} dx \right\}$ $= \frac{\pi^2}{64} - \left\{ \left[x \tan^{-1}(x) \right]_{-1}^0 - \left[\frac{1}{2} \ln 1+x^2 \right]_{-1}^0 \right\}$ $= \frac{\pi^2}{64} + \frac{\pi}{4} - \frac{1}{2} \ln 2$ OR Area of region $R = \frac{1}{2} \left(1 + \frac{\pi}{8} + 1 \right) \left(\frac{\pi}{4} \right) - \int_{-\frac{\pi}{4}}^0 (0 - \tan(y)) dy$ $= \frac{\pi^2}{64} + \frac{\pi}{4} + \left[\ln \sec y \right]_{-\frac{\pi}{4}}^0$ $= \frac{\pi^2}{64} + \frac{\pi}{4} + (0 - \ln \sqrt{2})$ $= \frac{\pi^2}{64} + \frac{\pi}{4} - \frac{1}{2} \ln 2$
8(ii) [2] [2]	After transformation, $y = -2\left(x + \frac{\pi}{8}\right) - 2 - \frac{\pi}{4}$ $= -2x - 2 - \frac{\pi}{4}$ After transformation, $y = \tan^{-1}\left(x + \frac{\pi}{8}\right)$ Volume $= \pi \int_{-\frac{\pi}{4}}^0 \left(-\frac{y}{2} - 1 - \frac{\pi}{4} \right)^2 - \left(\tan y - \frac{\pi}{8} \right)^2 dy$ $= 4.35$
9 [4]	$\frac{dy}{dx} + 2y = (x+1)e^{-2x}$ --- (1) Let $z = ye^{2x} \Rightarrow \frac{dz}{dx} = \frac{dy}{dx}e^{2x} + 2e^{2x}y$ Sub. into (1), $\frac{dz}{dx} + 2y = (x+1)e^{-2x}$ $\Rightarrow e^{2x} \frac{dy}{dx} + 2ye^{2x} = x+1$ $\Rightarrow \frac{dz}{dx} = x+1$ $\Rightarrow z = \int (x+1) dx$ $\Rightarrow z = \frac{x^2}{2} + x + c$ $\Rightarrow y = e^{-2x} \left(\frac{x^2}{2} + x + c \right)$

9(i) [1]	When $x = 0, y = 1, \therefore c = 1$. The particular solution is $y = e^{-2x} \left(\frac{x^2}{2} + x + 1 \right)$.
9(ii) [4]	$\frac{dy}{dx} + 2y = (x+1)e^{-2x}$ $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = e^{-2x} - 2(x+1)e^{-2x} = -(2x+1)e^{-2x}$ $\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} = -2e^{-2x} + 2(2x+1)e^{-2x} = 4xe^{-2x}$ When $x = 0, y = 1, \frac{dy}{dx} = -1, \frac{d^2y}{dx^2} = 1, \frac{d^3y}{dx^3} = -2$ Using Maclaurin's expansion, $y = 1 - x + \frac{1}{2!}x^2 - \frac{2}{3!}x^3 + \dots$ $= 1 - x + \frac{1}{2}x^2 - \frac{1}{3}x^3 + \dots$
9iii [2]	Replace x by $-2x$ in the standard series expansion of e^x and perform an expansion for $e^{-2x} \left(\frac{x^2}{2} + x + 1 \right)$ up to and including the term in x^3 . Compare the coefficients of this series with that of the expansion in (ii) to verify the correctness.
10a [7]	$V = \frac{1}{3}\pi r^2 h = 120 \Rightarrow h = \frac{360}{\pi r^2}$ From diagram, $r^2 + h^2 = l^2$ $S = \pi r l$ $S^2 = \pi^2 r^2 l^2$ $= \pi^2 r^2 (r^2 + h^2)$ $= \pi^2 r^2 \left(r^2 + \frac{360^2}{\pi^2 r^4} \right)$ $= \pi^2 r^4 + \frac{129600}{r^2}$ Differentiate w.r.t. r, we get $2S \frac{dS}{dr} = 4\pi^2 r^3 - \frac{259200}{r^3}$ For stationary values of S, set $\frac{dS}{dr} = 0$ $4\pi^2 r^3 = \frac{259200}{r^3} \Rightarrow r^6 = \frac{64800}{\pi^2}$ $\Rightarrow r = \frac{\sqrt[6]{64800}}{\sqrt[3]{\pi}} = 4.33 \text{ (3 s.f.) and } h = \frac{360}{\pi r^2} = 6.11 \text{ (3 s.f.)}$
10b i [3]	$\frac{dy}{dx} = -\frac{3\cos t}{5\sin t}$ <u>Equation of l</u>

	$y - 3\sin t = \frac{5\sin t}{3\cos t}(x - 5\cos t)$ $y = \frac{5}{3}x \tan t - \frac{25}{3}\sin t + 3\sin t$ $\text{or } y = \frac{5}{3}x \tan t - \frac{16}{3}\sin t$
10b ii [4]	At A, $y = 0 \therefore x = \frac{16}{5}\cos t$ At B, $x = 0 \therefore y = -\frac{16}{3}\sin t$ Mid-point of AB, M has coordinates $\left(\frac{8}{5}\cos t, -\frac{8}{3}\sin t\right)$ Let $x = \frac{8}{5}\cos t$ and $y = -\frac{8}{3}\sin t$ Then $\frac{x^2}{\left(\frac{8}{5}\right)^2} + \frac{y^2}{\left(\frac{8}{3}\right)^2} = 1$ $\Leftrightarrow 25x^2 + 9y^2 = 64$ 
11a i [2]	
11a ii [2]	$\int_0^a h(x) dx = \frac{63}{2}$ $\int_{-1}^a h(x) dx = \frac{63}{2} + \frac{1}{2} = 32$ $\int_{-1}^a h(x) dx = 4(8)$ $a = -1 + 4(6) = 23$

11b i [2]	 $y = a(x-2)^2 + 3$
11b ii [3]	In order for $g \circ k$ to exist, $R_k \subseteq D_g$ Domain of g is $(-\infty, 5) \cup (5, \infty)$ $k(x) = a(x-2)^2 + 3, x \geq 7$ Range of k is $[25a + 3, \infty)$ So, $25a + 3 > 5$ $a > \frac{2}{25}$ or 0.08
11b iii [3]	 Intersection $x=5.5917041$ $y=9.4501693$ When $f(x) > g(x)$, $x < 5$ or $x > 5.59$ (3 s.f.) Replace x with $-x$, $x > -5$ or $x < -5.59$ (3 s.f.)
12(a) [4]	$zz^* = \left \left(\sqrt{3} + i \right)^{\frac{2}{7}} \right ^2 = \left \sqrt{3} + i \right ^{\frac{4}{7}}$ $= 2^{\frac{4}{7}}$ $\arg z = \arg \left(\left(\sqrt{3} + i \right)^{\frac{2}{7}} \right)$ $= \frac{2}{7} \arg(\sqrt{3} + i)$ $= \frac{2}{7} \left(\frac{\pi}{6} \right) = \frac{\pi}{21}$

(b) [5]	$\arg\left(iw - \frac{i}{\sqrt{3}} + 1\right) = -\frac{\pi}{6}$ $\arg\left(i\left(w - \frac{1}{\sqrt{3}} - i\right)\right) = -\frac{\pi}{6}$ $\arg\left(w - \left(\frac{\sqrt{3}}{3} + i\right)\right) = -\frac{\pi}{2} - \frac{\pi}{6} = -\frac{2\pi}{3}$ 
12b i [2]	Least value of $\left z - \frac{1}{\sqrt{3}} - i\right$ $= \frac{1}{\sqrt{3}} - (-\sqrt{3}) - 2$ $= \frac{4\sqrt{3}}{3} - 2$
12b ii [2]	$\frac{2\pi}{3} \leq \arg\left(z - \frac{1}{\sqrt{3}} - i\right) \leq \pi \text{ or } -\pi < \arg\left(z - \frac{1}{\sqrt{3}} - i\right) \leq -\frac{2\pi}{3}$